# CSE 114A: Fall 2023

# Foundations of Programming Languages

*Lexing and Parsing*

Owen Arden
UC Santa Cruz

*Based on course materials developed by Nadia Polikarpova*

## Plan for this week

**Last week:**

- How do we *evaluate* a program given its AST?

```
eval :: Env -> Expr -> Value
```

**This week:**

- How do we *convert* program text into an AST?

```
parse :: String -> Expr
```



## Example: calculator with vars

AST representation:

```
data Aexpr
  = AConst  Int
  | AVar    Id
  | APlus   Aexpr Aexpr
  | AMinus  Aexpr Aexpr
  | AMul    Aexpr Aexpr
  | ADiv    Aexpr Aexpr
```

Evaluator:

```
eval :: Env -> Aexpr -> Value
...
```

# Example: calculator with vars

Using the evaluator:

```
λ> eval [] (APlus (AConst 2) (AConst 6))
8

λ> eval [("x", 16), ("y", 10)] (AMinus (AVar "x") (AVar "y"))
6

λ> eval [("x", 16), ("y", 10)] (AMinus (AVar "x") (AVar "z"))
*** Exception: Error {errMsg = "Unbound variable z”}
```

But writing ASTs explicitly is really tedious, we are used to writing programs as text!



# Example: calculator with vars

We want to write a function that converts strings to ASTs if possible:

```
parse :: String -> Aexpr
```

For example:

```
λ> parse "2 + 6"
APlus (AConst 2) (AConst 6)

λ> parse "(x - y) / 2"
ADiv (AMinus (AVar "x") (AVar "y")) (AConst 2)

λ> parse "2 +"
*** Exception: Error {errMsg = "Syntax error"}
```



# Two-step-strategy

How do I read a sentence “He ate a bagel”?

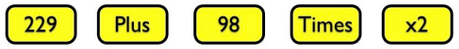
- First split into words: `["He", "ate", "a", "bagel"]`
- Then relate words to each other: “He” is the subject, “ate” is the verb, etc

Let’s do the same thing to “read” programs!

# 1. Lexing: From String to Tokens

A string is a list of *characters*:

First we aggregate characters that "belong together" into **tokens** (i.e. the "words" of the program):




We distinguish tokens of different kinds based on their format:

- all numbers: integer constant
- alphanumeric, starts with a letter: identifier
- +: plus operator
- etc



# 2. Parsing: From Tokens to AST

Next, we convert a sequence of tokens into an AST

- This is hard…
- … but the hard parts do not depend on the language!

**Parser generators**

- Given the description of the *token format* generates a *lexer*
- Given the description of the *grammar* generates a *parser*

We will be using parser generators, so we only care about how to describe the token format and the grammar



# Lexing

We will use the tool called `alex` to generate the **lexer**

Input to `alex`: a `.x` file that describes the *token format*

# Tokens

First we list the kinds of tokens we have in the language:

```
data Token
  = NUM    AlexPosn Int
  | ID     AlexPosn String
  | PLUS   AlexPosn
  | MINUS  AlexPosn
  | MUL    AlexPosn
  | DIV    AlexPosn
  | LPAREN AlexPosn
  | RPAREN AlexPosn
  | EOF    AlexPosn
```



# Token rules

Next we describe the format of each kind of token using a rule:

```
  [\+]                        { \p _ -> PLUS   p }
  [\-]                        { \p _ -> MINUS  p }
  [\*]                        { \p _ -> MUL    p }
  [\/]                        { \p _ -> DIV    p }
  \(                          { \p _ -> LPAREN p }
  \)                          { \p _ -> RPAREN p }
  $alpha [$alpha $digit \_ \']* { \p s -> ID     p s }
  $digit+                     { \p s -> NUM p (read s) }
```

Each line consist of:

- a *regular expression* that describes which strings should be recognized as this token
- a Haskell expression that generates the token

You read it as:

- if at position p in the input string
- you encounter a substring s that matches the *regular expression*
- evaluate the Haskell expression with arguments p and s



# Regular Expressions

A regular expression has one of the following forms:

- `[c1 c2 ... cn]` matches *any of* the characters `c1 .. cn`
  - `[0-9]` matches *any digit*
  - `[a-z]` matches *any lower-case letter*
  - `[A-Z]` matches *any upper-case letter*
  - `[a-z A-Z]` matches *any letter*
- `R1 R2` matches a string `s1 + + s2` where `s1` matches `R1` and `s2` matches `R2`
  - e.g. `[0-9] [0-9]` matches any two-digit string
- `R+` matches *one or more* repetitions of what `R` matches
  - e.g. `[0-9]+` matches a natural number
- `R*` matches *zero or more* repetitions of what `R` matches

# QUIZ

Which of the following strings are matched by [a-z A-Z] [a-z A-Z 0-9]*? *

- (A) (empty string)
- (B) 5
- (C) x5
- (D) x
- (E) C and D

**http://tiny.cc/cmps112-regex-ind**



# QUIZ

Which of the following strings are matched by [a-z A-Z] [a-z A-Z 0-9]*? *

- (A) (empty string)
- (B) 5
- (C) x5
- (D) x
- (E) C and D

**http://tiny.cc/cmps112-regex-grp**



# Back to token rules

We can **name** some common regexps like:

```
$digit = [0-9]
$alpha = [a-z A-Z]
```

and write `[a-z A-Z] [a-z A-Z 0-9]*` as `$alpha [$alpha $digit]*`

```
  [\+]                          { \p _ -> PLUS   p }
  [\-]                          { \p _ -> MINUS  p }
  [\*]                          { \p _ -> MUL    p }
  [\/]                          { \p _ -> DIV    p }
  \(                            { \p _ -> LPAREN p }
  \)                            { \p _ -> RPAREN p }
  $alpha [$alpha $digit \_ \']* { \p s -> ID     p s }
  $digit+                       { \p s -> NUM p (read s) }
```

- When you encounter a `+`, generate a `PLUS` token … etc
- When you encounter a nonempty string of digits, convert it into an integer and generate a `NUM`
- When you encounter an alphanumeric string that starts with a letter, save it in an `ID` token

# Running the Lexer

From the token rules, `alex` generates a function `alexScan` which

- given an input string, find the *longest* prefix `p` that matches one of the rules
- if `p` is empty, it fails
- otherwise, it converts `p` into a token and returns the rest of the string

We wrap this function into a handy function

```
parseTokens :: String -> Either ErrMsg [Token]
```

which repeatedly calls `alexScan` until it consumes the whole input string or fails



# Running the Lexer

We can test the function like so:

```
λ> parseTokens "23 + 4 / off -"
Right [ NUM (AlexPn 0 1 1) 23
      , PLUS (AlexPn 3 1 4)
      , NUM (AlexPn 5 1 6) 4
      , DIV (AlexPn 7 1 8)
      , ID (AlexPn 9 1 10) "off"
      , MINUS (AlexPn 13 1 14)
      ]
λ> parseTokens "%"
Left "lexical error at 1 line, 1 column"
```



# QUIZ

What is the result of parseTokens "92zoo" (positions omitted for readability)? *

- (A) Lexical error
- (B) [ID "92zoo"]
- (C) [NUM "92"]
- (D) [NUM "92", ID "zoo"]

**http://tiny.cc/cmps112-ptoken-ind**

# QUIZ

What is the result of parseTokens "92zoo" (positions omitted for readability)? *

- (A) Lexical error
- (B) [ID "92zoo"]
- (C) [NUM "92"]
- (D) [NUM "92", ID "zoo"]

http://tiny.cc/cmps112-ptoken-grp



# Parsing

We will use the tool called `happy` to generate the **parser**

Input to `happy`: a `.y` file that describes the *grammar*



# Parsing

Wait, wasn't this the grammar?

```
data Aexpr
  = AConst  Int
  | AVar    Id
  | APlus   Aexpr Aexpr
  | AMinus  Aexpr Aexpr
  | AMul    Aexpr Aexpr
  | ADiv    Aexpr Aexpr
```

This was *abstract syntax*

Now we need to describe *concrete syntax*

- What programs look like when written as text
- and how to map that text into the abstract syntax

# Grammars

A grammar is a recursive definition of a set of trees

- each tree is a *parse tree* for some string
- *parse* a string `s` = find a parse tree for `s` that belongs to the grammar

A grammar is made of:

- **Terminals**: the leaves of the tree (tokens!)
- **Nonterminals:** the internal nodes of the tree
- **Production Rules** that describe how to "produce" a non-terminal from terminals and other non-terminals
  - i.e. what children each nonterminal can have:

```
Aexpr :   -- NT Aexpr can have as children:
  | Aexpr '+' Aexpr  { ... } -- NT Aexpr, T '+', and NT Aexpr, or
  | Aexpr '-' AExpr  { ... } -- NT Aexpr, T '-', and NT Aexpr, or
  | ...
```



# Terminals

Terminals correspond to the *tokens* returned by the lexer

In the `.y` file, we have to declare with terminals in the rules correspond to which tokens from the `Token` datatype:

```
%token
    TNUM  { NUM _ $$ }
    ID    { ID _ $$  }
    '+'   { PLUS _   }
    '-'   { MINUS _  }
    '*'   { MUL _    }
    '/'   { DIV _    }
    '('   { LPAREN _ }
    ')'   { RPAREN _ }
```

- Each thing on the left is terminal (as appears in the production rules)
- Each thing on the right is a Haskell pattern for datatype `Token`
- We use `$$` to designate one parameter of a token constructor as the token **value**
  - we will refer back to it from the production rules



# Production rules

Next we define productions for our language:

```
Aexpr : TNUM                    { AConst $1    }
      | ID                      { AVar   $1    }
      | '(' Aexpr ')'           { $2           }
      | Aexpr '*' Aexpr         { AMul   $1 $3 }
      | Aexpr '+' Aexpr         { APlus  $1 $3 }
      | Aexpr '-' Aexpr         { AMinus $1 $3 }
```

The expression on the right computes the *value* of this node

- `$1` `$2` `$3` refer to the *values* of the respective child nodes

# Production rules

**Example:** parsing `(2)` as `AExpr`:

1. Lexer returns a sequence of `Tokens`: `[LPAREN, NUM 2, RPAREN]`
2. `LPAREN` is the token for terminal `'('`, so let's pick production `'(' Aexpr ')'`
3. Now we have to parse `NUM 2` as `Aexpr` and `RPAREN` as `')'`
4. `NUM 2` is a token for nonterminal `TNUM`, so let's pick production `TNUM`
5. The value of this `Aexpr` node is `AConst 2`, since the value of `TNUM` is `2`
6. The value of the top-level `Aexpr` node is also `AConst 2` (see the `'(' Aexpr ')'`production)



# QUIZ

What is the value of the root AExpr node when parsing 1 + 2 + 3?
*

```
Aexpr : TNUM                  { AConst $1    }
      | ID                    { AVar   $1    }
      | '(' Aexpr ')'         { $2           ]
      | Aexpr '*' Aexpr       { AMul   $1 $3 ]
      | Aexpr '+' Aexpr       { APlus  $1 $3 ]
      | Aexpr '-' Aexpr       { AMinus $1 $3 ]
```

- (A) Cannot be parsed as AExpr
- (B) 6
- (C) APlus (APlus (AConst 1) (AConst 2)) (AConst 3)
- (D) APlus (AConst 1) (APlus (AConst 2) (AConst 3))

**http://tiny.cc/cmps112-aexpr-ind**



# QUIZ

What is the value of the root AExpr node when parsing 1 + 2 + 3?
*

```
Aexpr : TNUM                  { AConst $1    }
      | ID                    { AVar   $1    }
      | '(' Aexpr ')'         { $2           ]
      | Aexpr '*' Aexpr       { AMul   $1 $3 ]
      | Aexpr '+' Aexpr       { APlus  $1 $3 ]
      | Aexpr '-' Aexpr       { AMinus $1 $3 ]
```

- (A) Cannot be parsed as AExpr
- (B) 6
- (C) APlus (APlus (AConst 1) (AConst 2)) (AConst 3)
- (D) APlus (AConst 1) (APlus (AConst 2) (AConst 3))

**http://tiny.cc/cmps112-aexpr-grp**

# Running the Parser

First, we should tell the parser that the top-level non-terminal is `AExpr`:

```
%name aexpr
```

From the production rules and this line, `happy` generates a function `aexpr` that tries to parse a sequence of tokens as `AExpr`

We package this function together with the lexer and the evaluator into a handy function

```
evalString :: Env -> String -> Int
```

We can test the function like so:

```
λ> evalString [] "1 + 3 + 6"
10
λ> evalString [("x", 100), ("y", 20)] "x - y"
80
λ> evalString [] "2 * 5 + 5"
20
λ> evalString [] "2 - 1 - 1"
2
```



# Precedence and associativity

```
λ> evalString [] "2 * 5 + 5"
20
```

The problem is that our grammar is **ambiguous**!

There are multiple ways of parsing the string `2 * 5 + 5`, namely

- `APlus (AMul (AConst 2) (AConst 5)) (AConst 5)` (good)
- `AMul (AConst 2) (APlus (AConst 5) (AConst 5))` (bad!)

*Wanted:* tell `happy` that `*` has higher **precedence** than `+`!



# Precedence and associativity

```
λ> evalString [] "2 - 1 - 1"
2
```

There are multiple ways of parsing `2 - 1 - 1`, namely

- `AMinus (AMinus (AConst 2) (AConst 1)) (AConst 1)` (good)
- `AMinus (AConst 2) (AMinus (AConst 1) (AConst 1))` (bad!)

*Wanted:* tell `happy` that `-` is **left-associative**!

How do we communicate precedence and associativity to `happy`?

# Solution 1: Grammar factoring

We can split the `AExpr` non-terminal into multiple "levels"

```
Aexpr : Aexpr '+' Aexpr2
      | Aexpr '-' Aexpr2
      | Aexpr2

Aexpr2 : Aexpr2 '*' Aexpr3
       | Aexpr2 '/' Aexpr3
       | Aexpr3

Aexpr3 : TNUM
       | ID
       | '(' Aexpr ')'
```

Intuition: `AExpr2` "binds tighter" than `AExpr`, and `AExpr3` is the tightest

Now I cannot parse the string `2 * 5 + 5` as

- `AMul (AConst 2) (APlus (AConst 5) (AConst 5))` **Why?**

Because the RHS of `*` has to be `AExpr3`, while `5 + 5` is *not* an `AExpr3` (it's an `AExpr`)



# Solution 2: Parser directives

This problem is so common that parser generators have a special syntax for it!

```
%left '+' '-'
%left '*' '/'
```

What this means:

- All our operators are left-associative
- Operators on the lower line have higher precedence